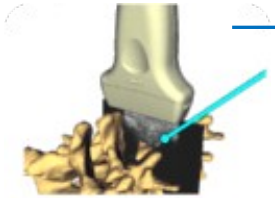
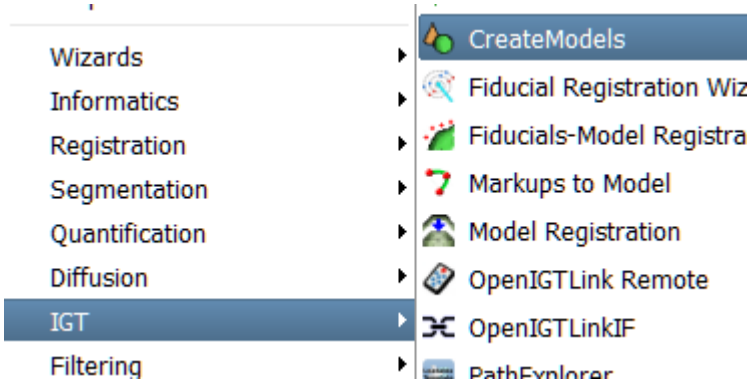
Tool models

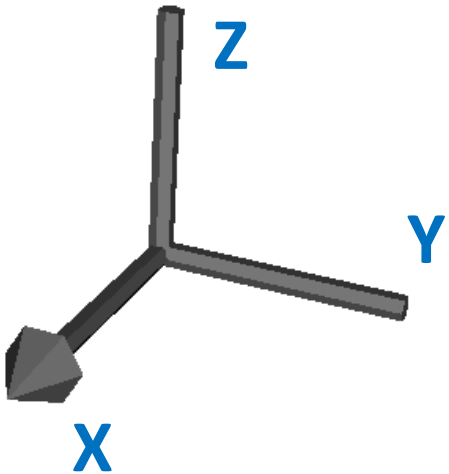


SlicerIGT Tutorial Series

Create simple models

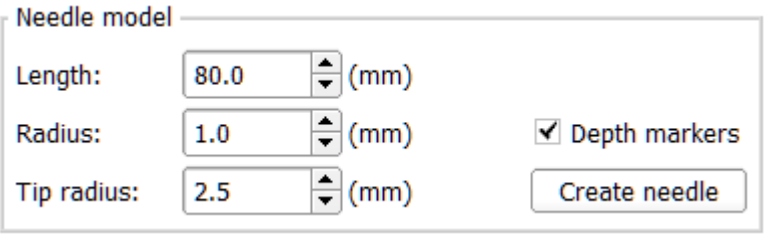
- Select **IGT / CreateModels** module
- Use the module widget on the left to create simple models with size parameters
- In the coordinate system model, X is marked with an arrow tip, Y and Z are defined if right-handedness is assumed

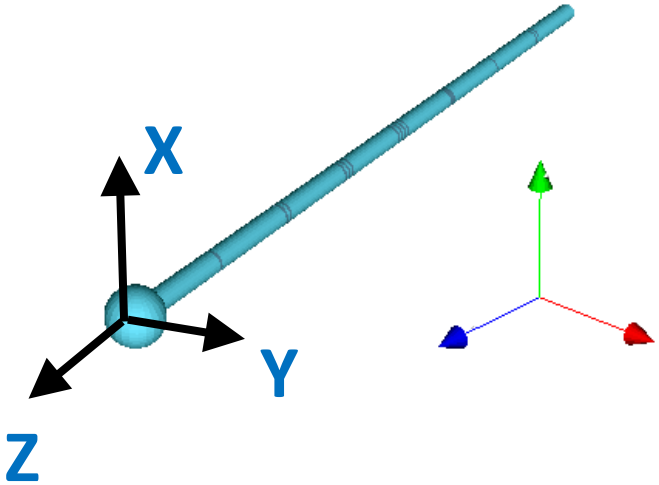




Needle / stylus models

- Optional depth markers can be created for the needle model. Marks are 10 mm apart as in most medical needles.
- Needle models and stylus models conventionally point towards the Z axis

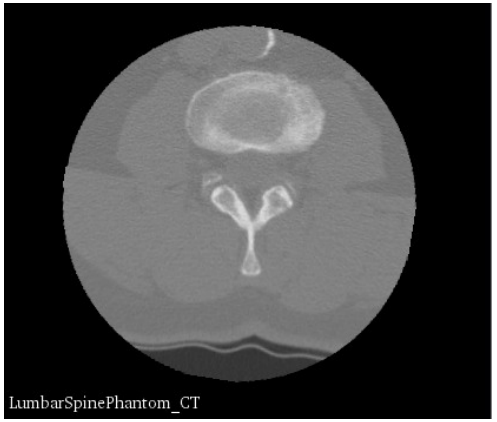


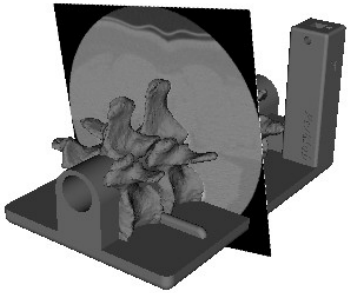


Load models from PLUS catalog

- Select IGT / Plus model catalog browser module
- Click Show/hide catalog to open the PLUS model catalog website
- Click on .stl files to download a model
- Additional files are also available, e.g. a CT volume of the spine model

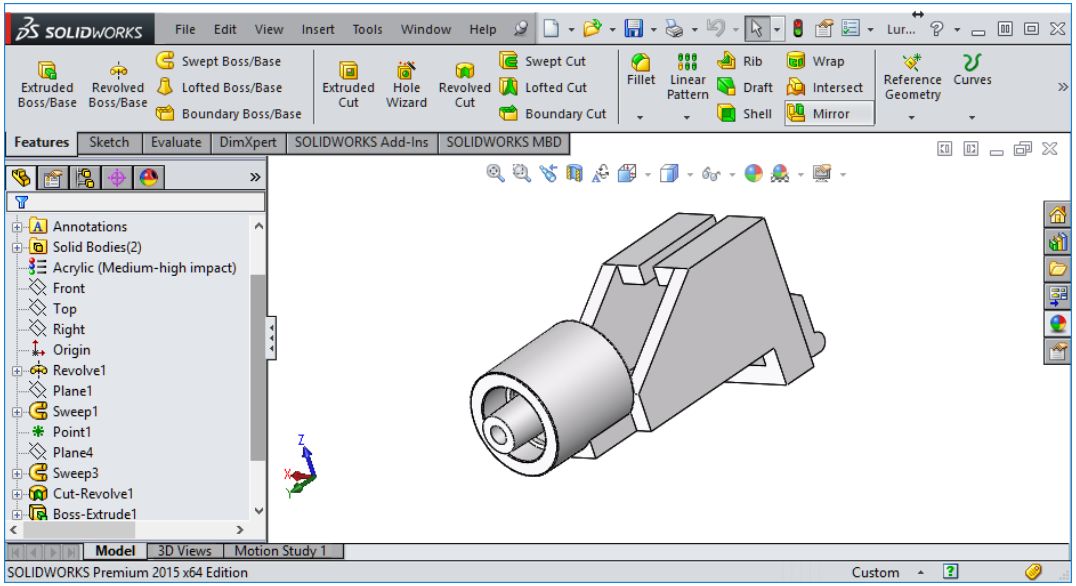
LumbarSpinePhantom.stl (rev.4795)
LumbarSpinePhantom_CT.mha (rev.4795)
Source file(s)





Create models in CAD software

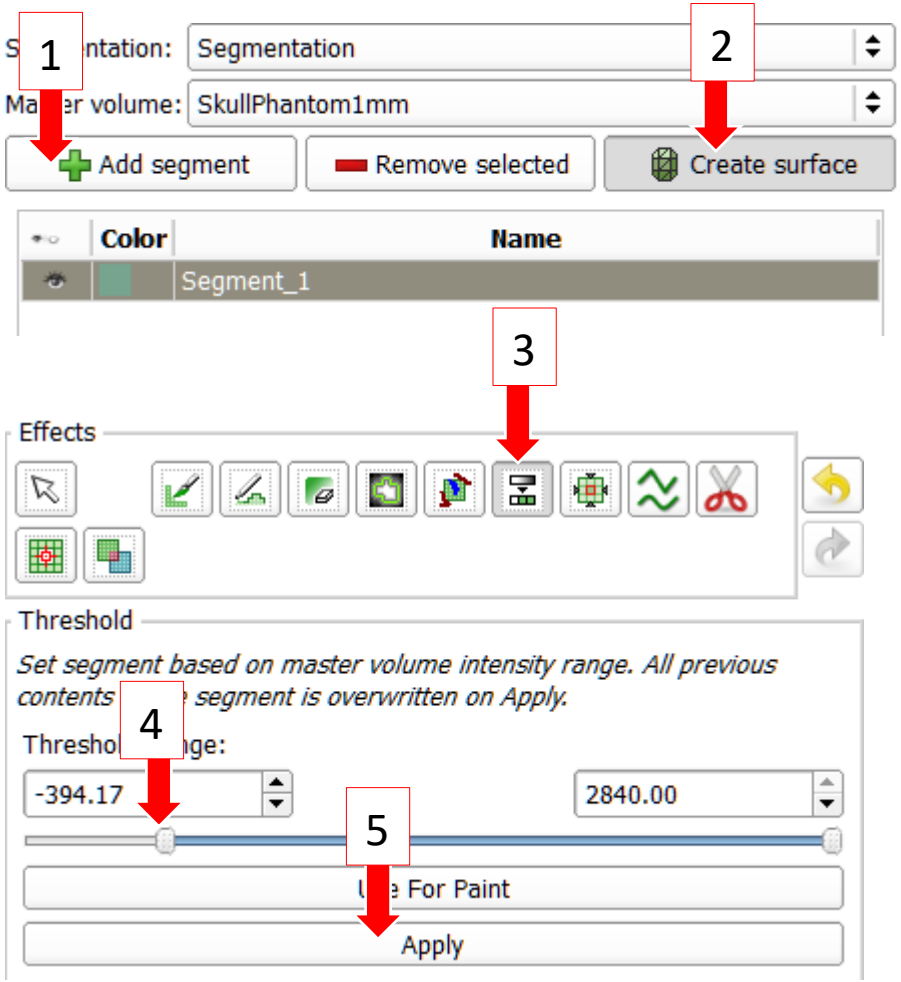
- FreeCAD, SolidWorks, Blender, etc.
- Save in STL format, units should be millimeters
- When exporting model in STL, choose and document an intuitive coordinate system

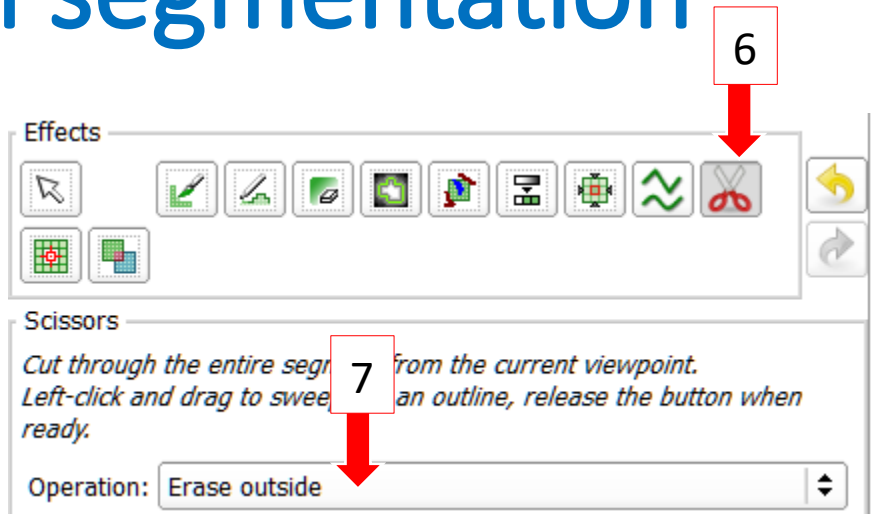


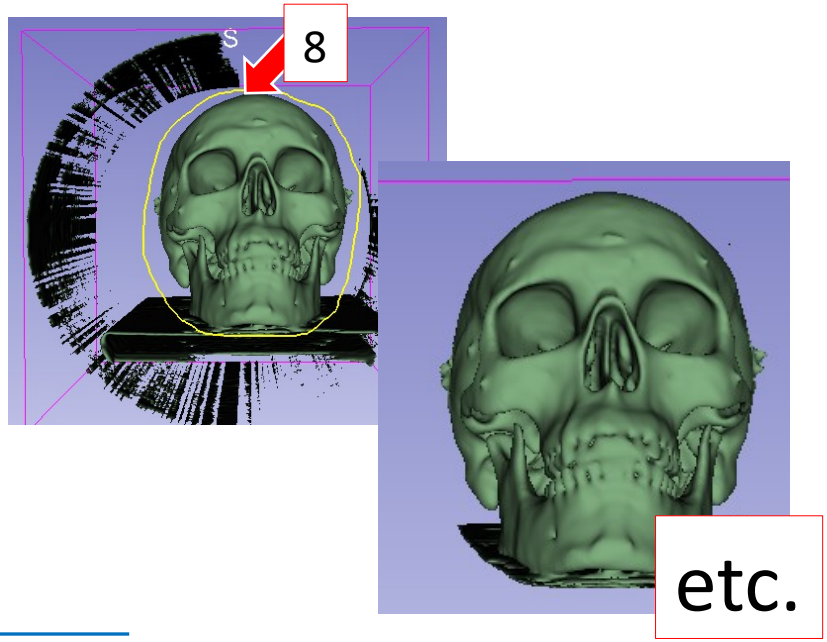
Models by image segmentation

- Load **SkullPhantom1mm.nrrd** from the tutorial data folder
- Select Segment Editor module
- Click Add segment
- Use the segmentation tools ("***Effects***") to segment
- Hint: Such objects CT scanned in air can be quicly segmented using threshold followed by scissors
- Select **Segmentations** module
- Export a new model under the **Export/import segments** panel

Skull model from segmentation



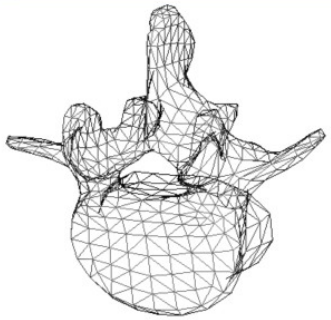




Models and Slicer model nodes

- Models are thin/hollow surfaces, most common objects in computer graphics
- Slicer wraps a **vtkMRMLModelNode** object around models, and keeps visualization properties in **vtkMRMLModelDisplayNode**
- https://www.slicer.org/wiki/Modules:Models-Documentation

surfaces



lines

points